Project: Worked examples
Title: Beam and 2D frame analysis
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Linear elastic analysis of a plane frame or continuous beam by the direct stiffness method: reactions, member end forces, axial force, shear, bending moment and deflection along every member with their extremes and positions, for each load combination and as an envelope. Global x to the right, y up; moments and couples anticlockwise positive; sagging moments positive along a member from node i to node j.

Three freedoms per node; Euler–Bernoulli member stiffness EA/L and EI/L³ terms; end releases by static condensation; supports by removing freedoms or adding spring stiffness; equivalent nodal loads from exact fixed-end forces; K u = F solved by banded LDL[1] after a reverse Cuthill–McKee ordering (a non-positive pivot is reported as a mechanism). Internal forces and deflections from the member end actions and Macaulay functions of the span loads; extremes at stations plus the zero-shear and zero-slope points found by bisection.

Ref: Hibbeler, Structural Analysis Ch. 16 — Plane frame analysis using the stiffness method
Ref: Roark's Formulas for Stress and Strain Ch. 8, Table 8.1 — Shear, moment, slope and deflection formulas for elastic straight beams
Ref: BS EN 1990 6.4.3 — Combinations of actions (user-entered factors)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Default modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
nodes	Nodes	4 row(s)		drawing: DRG-PF-010 rev B	any
elements	Elements	3 row(s)		reference: UC 203×203×46 columns, UB 305×165×40 rafter (Blue Book)	any
loads	Loads	2 row(s)		spec: Load schedule LS-02: roof 15 kN/m; wind 24 kN at eaves	any
combinations	Combinations (blank = each case alone)	3 row(s)		assumption: Characteristic combinations for the worked example.	any
report	Combination to report in detail (blank = first)	G+W		assumption: Gravity plus wind reported.	any
n_div	Stations per member	16		assumption: 16 stations per member.	4 … 100

3. Assumptions

A1. Linear elastic, small displacements, first order; no shear deformation, no P–Δ, no buckling. (Euler–Bernoulli direct stiffness.)
A2. Members are prismatic between nodes and connect rigidly unless a release is stated; a release

transmits no moment at that end. (Static condensation of the released rotation.)
A3. Distributed loads act perpendicular to the member unless marked as gravity loads, which act in global −y per metre of member length. (Load table convention.)
A4. combinations (Combinations (blank = each case alone)) is taken as 3 row(s). (Characteristic combinations for the worked example.)
A5. report (Combination to report in detail (blank = first)) is taken as G+W. (Gravity plus wind reported.)
A6. n_div (Stations per member) is taken as 16. (16 stations per member.)

4. Working

```
[4] n_nodes = nodes
= 4
= 4
```
Nodes in the model
```
[5] n_elements = elements
= 3
= 3
```
Members in the model
```
[6] n_comb = combinations
= G+W, G
= 2
```
Load combinations analysed
```
[7] comb = reported combination
= G+W
= "G+W"
```
The combination detailed below
```
[8] R1_x = reaction at node 1, x
= Σ member end forces − applied nodal loads
= -2.9968 kN
```
Horizontal reaction at node 1
```
[9] R1_y = reaction at node 1, y
= Σ member end forces − applied nodal loads
= 45 kN
```
Vertical reaction at node 1
```
[10] R1_m = reaction at node 1, moment
= Σ member end moments − applied nodal couple
= -1.4552e-14 kN·m
```
Moment reaction at node 1
```
[11] R4_x = reaction at node 4, x
= Σ member end forces − applied nodal loads
= -21.003 kN
```
Horizontal reaction at node 4
```
[12] R4_y = reaction at node 4, y
= Σ member end forces − applied nodal loads
= 75 kN
```
Vertical reaction at node 4
```
[13] R4_m = reaction at node 4, moment
= Σ member end moments − applied nodal couple
= -2.9104e-14 kN·m
```
Moment reaction at node 4
```
[14] sum_R_y = Σ R_y
= all supports
= 120 kN
```
Total vertical reaction (equilibrium check against the applied load)

[15] sum_R_x = Σ R_x
= all supports
= -24 kN
Total horizontal reaction
[16] M_max_e1 = member 1: max sagging M
= at x = 5 m from node 1
= 14.984 kN·m
Member 1 maximum moment
[17] M_min_e1 = member 1: min (hogging) M
= at x = 0 m
= 0 kN·m
Member 1 minimum moment
[18] V_e1 = member 1: max |V|
= 2.9968 kN
= 2.9968 kN
Member 1 maximum shear
[19] N_e1 = member 1: max |N| (tension +)
= -45 kN
= -45 kN
Member 1 axial force of greatest magnitude
[20] d_e1 = member 1: max |deflection|
= at x = 5 m
= 74.678 mm
Member 1 maximum transverse deflection
[21] M_max_e2 = member 2: max sagging M
= at x = 3 m from node 2
= 82.484 kN·m
Member 2 maximum moment
[22] M_min_e2 = member 2: min (hogging) M
= at x = 8 m
= -105.02 kN·m
Member 2 minimum moment
[23] V_e2 = member 2: max |V|
= 75 kN
= 75 kN
Member 2 maximum shear
[24] N_e2 = member 2: max |N| (tension +)
= -21.003 kN
= -21.003 kN
Member 2 axial force of greatest magnitude
[25] d_e2 = member 2: max |deflection|
= at x = 3.4916 m
= 25.45 mm
Member 2 maximum transverse deflection
[26] M_max_e3 = member 3: max sagging M
= at x = 5 m from node 3
= 0 kN·m
Member 3 maximum moment
[27] M_min_e3 = member 3: min (hogging) M
= at x = 0 m
= -105.02 kN·m
Member 3 minimum moment

[28] V_e3 = member 3: max |V|
= 21.003 kN
= 21.003 kN
Member 3 maximum shear
[29] N_e3 = member 3: max |N| (tension +)
= -75 kN
= -75 kN
Member 3 axial force of greatest magnitude
[30] d_e3 = member 3: max |deflection|
= at x = 0.31515 m
= 75.042 mm
Member 3 maximum transverse deflection
[31] M_max = max sagging moment
= member 2 at x = 3 m
= 82.484 kN·m
Maximum moment in the reported combination
[32] x_Mmax = position of M_max
= from node i of member 2
= 3 m
Position of the maximum moment
[33] M_min = min (hogging) moment
= member 3 at x = 0 m
= -105.02 kN·m
Minimum moment in the reported combination
[34] V_max = max |V|
= member 2
= 75 kN
Maximum shear in the reported combination
[35] N_max = max tension
= member 2
= -21.003 kN
Maximum axial tension
[36] N_min = max compression (negative)
= member 3
= -75 kN
Maximum axial compression
[37] d_max = max |deflection|
= member 3 at x = 0.31515 m
= 75.042 mm
Maximum transverse deflection
[38] M_max_env = envelope max M
= G+W, member 2
= 82.484 kN·m
Maximum moment over all combinations
[39] M_min_env = envelope min M
= G+W, member 3
= -105.02 kN·m
Minimum moment over all combinations
[40] V_max_env = envelope max |V|
= G+W, member 2
= 75 kN
Maximum shear over all combinations

[41] d_max_env = envelope max |□|
= G+W, member 3
= 75.042 mm
Maximum deflection over all combinations

5. Results

SYMBOL	VALUE	DESCRIPTION
R1_x	-2.9968 kN	R_x node 1
R1_y	45 kN	R_y node 1
R1_m	-1.4552e-14 kN·m	M node 1
R4_x	-21.003 kN	R_x node 4
R4_y	75 kN	R_y node 4
R4_m	-2.9104e-14 kN·m	M node 4
sum_R_y	120 kN	Σ R_y
sum_R_x	-24 kN	Σ R_x
M_max_e1	14.984 kN·m	M max, member 1
M_min_e1	0 kN·m	M min, member 1
V_e1	2.9968 kN	|V| max, member 1
N_e1	-45 kN	N, member 1
d_e1	74.678 mm	δ max, member 1
M_max_e2	82.484 kN·m	M max, member 2
M_min_e2	-105.02 kN·m	M min, member 2
V_e2	75 kN	|V| max, member 2
N_e2	-21.003 kN	N, member 2
d_e2	25.45 mm	δ max, member 2
M_max_e3	0 kN·m	M max, member 3
M_min_e3	-105.02 kN·m	M min, member 3
V_e3	21.003 kN	|V| max, member 3
N_e3	-75 kN	N, member 3
d_e3	75.042 mm	δ max, member 3
M_max	82.484 kN·m	M max
x_Mmax	3 m	x of M max
M_min	-105.02 kN·m	M min
V_max	75 kN	|V| max
N_max	-21.003 kN	N max (tension)
N_min	-75 kN	N min (compression)
d_max	75.042 mm	δ max
M_max_env	82.484 kN·m	M max, envelope

SYMBOL	VALUE	DESCRIPTION
M_min_env	-105.02 kN·m	M min, envelope
V_max_env	75 kN	|V| max, envelope
d_max_env	75.042 mm	δ max, envelope

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

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Template struct.frame-2d@1.0.0 Content hash 62fe68544f250603 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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